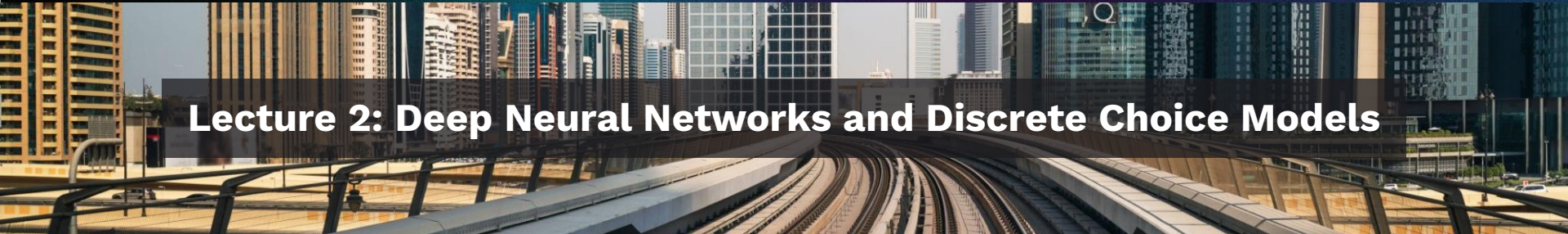


AI for Transportation: From concepts to implementation



Lecture 2: Deep Neural Networks and Discrete Choice Models

AI for Transportation

L1: Introduction

L2: DNNs and Discrete Choice Models

L3: Real-World AI Case Studies

L4: Generative AI

AI for Transportation

L2: DNNs and Discrete Choice Models

S1: Background

S2: Extracting Economic Information for Interpretation

S3: Architectural Design with Alternative-Specific Utility Function

S4: Theory-Based Residual Neural Network

S5: Deep hybrid model with urban imagery

Predicting Individual Choices with Discrete Choice Models

Travel mode choice



Discrete choices in urban transportation

Travel mode choice



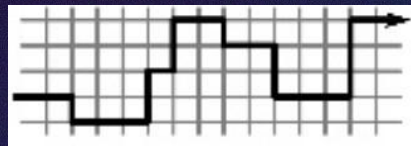
Uber single passenger vs. ride-sharing

Residential & job location choice

Auto ownership

Destination choice

Route choice



Aggregate vs. disaggregate demand analysis

Aggregate demand analysis:

- Urban grids

- Census blocks, census tracts, cities, etc.

Individual demand analysis (discrete choice):

- Individuals/households

Relationship

- From individual to aggregate: easy

- From aggregate to individual: difficult

Taxi Demand Prediction



Classical analytical framework for discrete choice

Decision-Maker

Individuals (person/household)

Socio-economic variables (e.g. age, gender, income)

Alternatives

Decision-maker i selects one and only one alternative from a choice set $C = \{1, 2, \dots, K\}$

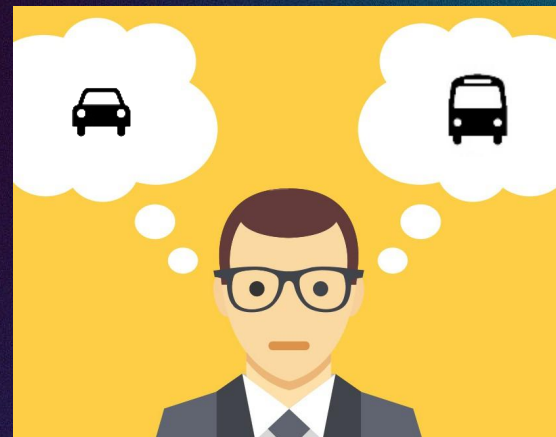
Collectively exhaustive & mutually exclusive

Attributes of alternatives

e.g. price, quality, time, reliability,...

Decision rule

Utility maximization, etc.



Random utility models

$$U_{in} = V_{in} + \varepsilon_{in}$$

More Specifically

$$U_{in} = V_{in}(X_i, S_n; \beta) + \varepsilon_{in}$$

Linear parameters function

$$U_{in} = \beta_1 x_{1in} + \beta_2 x_{2in} + \beta_3 x_{3in} + \dots + \varepsilon_{in}$$

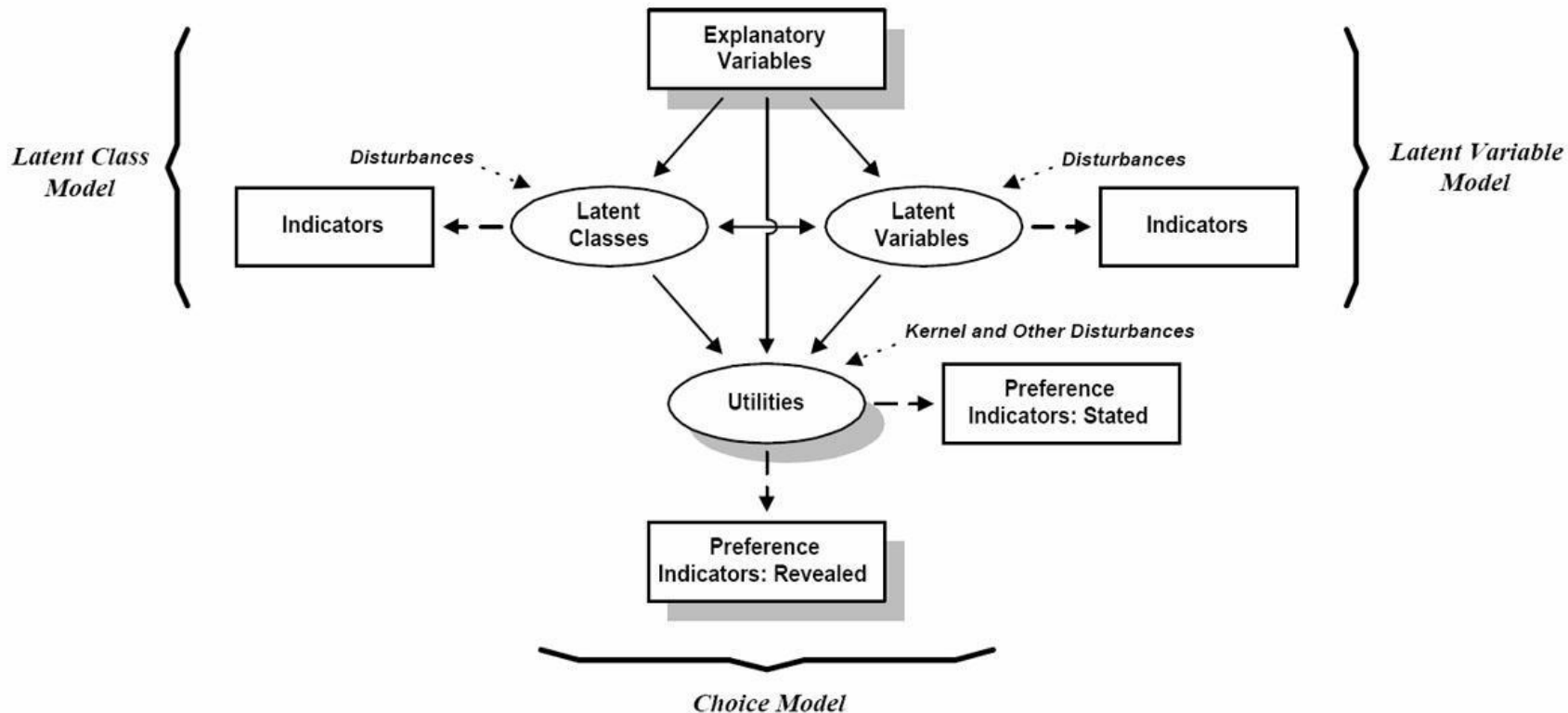
Mathematical formulation

Probability

$$\begin{aligned} P(y_i | x_i) &= f(x_i)P(i | C_n) = P[U_{in} \geq U_{jn} \quad \forall j \in C_n] \\ &= P\left[U_{in} = \max_{j \in C_n} U_{jn}\right] \end{aligned}$$

Multinomial Logit

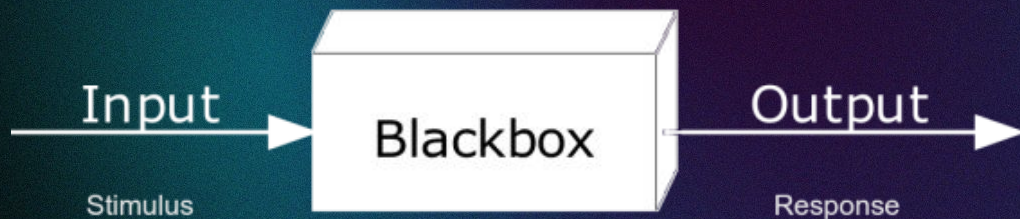
$$P(i | C_n) = \frac{e^{V_{in}}}{\sum_{j \in C_n} e^{V_{jn}}}$$



Predicting Individual Choices with Deep Neural Network

Treat choice analysis as a prediction task

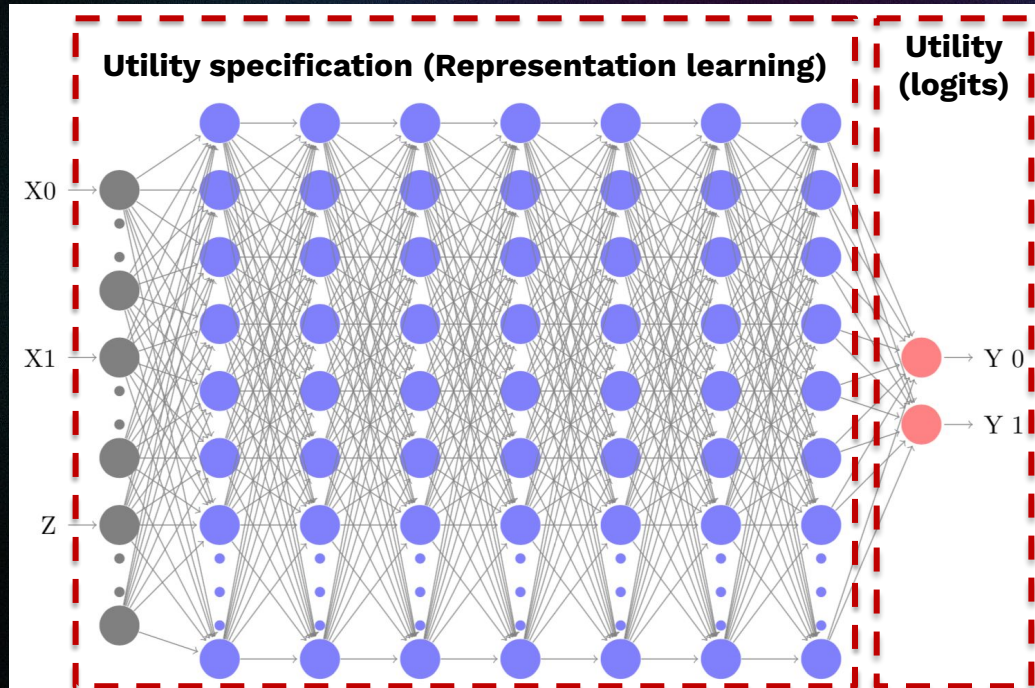
$$f : x \mapsto y, \quad y \in \{1, 2, \dots, K\}$$



Use any predictive ML classifier for choice analysis

- Deep neural networks (any architecture, hyper parameters, etc.)
- Discriminant analysis
- Bayesian model (Naïve Bayesian, etc.)
- Support vector machine (any kernel)
- K nearest neighbors (KNN)
- Decision trees
- Random forests

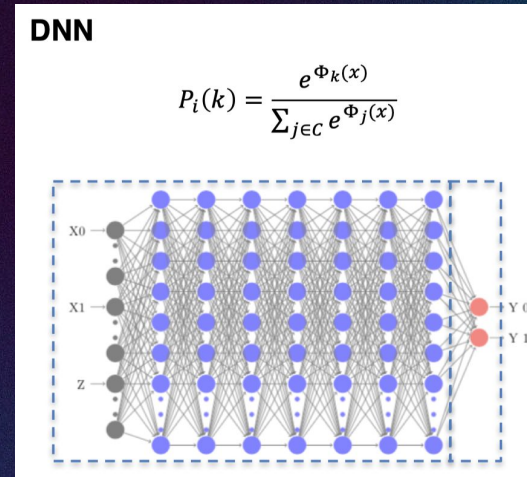
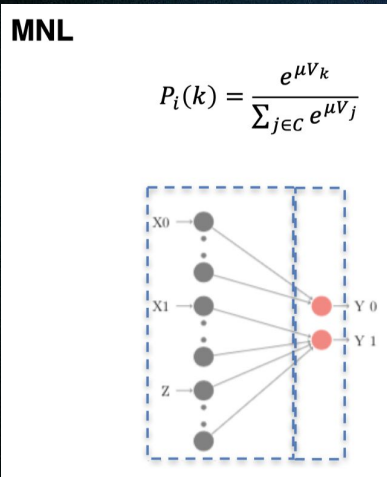
Deep neural networks for Choice Analysis



DNN for Binary Choice

Connecting Discrete Choice Model with Deep Neural Network

Comparing MNL and DNN: MNL and DNN are similar



1. Utility specification & comparison exist in both MNL and DNN
2. McFadden (1974) provided proof of the RUM and softmax activation function

Extracting Economic Information for Interpretation

AI for Transportation

Wang, S., Wang, Q., & Zhao, J. (2020). Deep neural networks for choice analysis: Extracting complete economic information for interpretation. *Transportation Research Part C* - <https://doi.org/10.1016/j.trc.2020.102701>

L1: DNNs and Discrete Choice Models

S1: Background

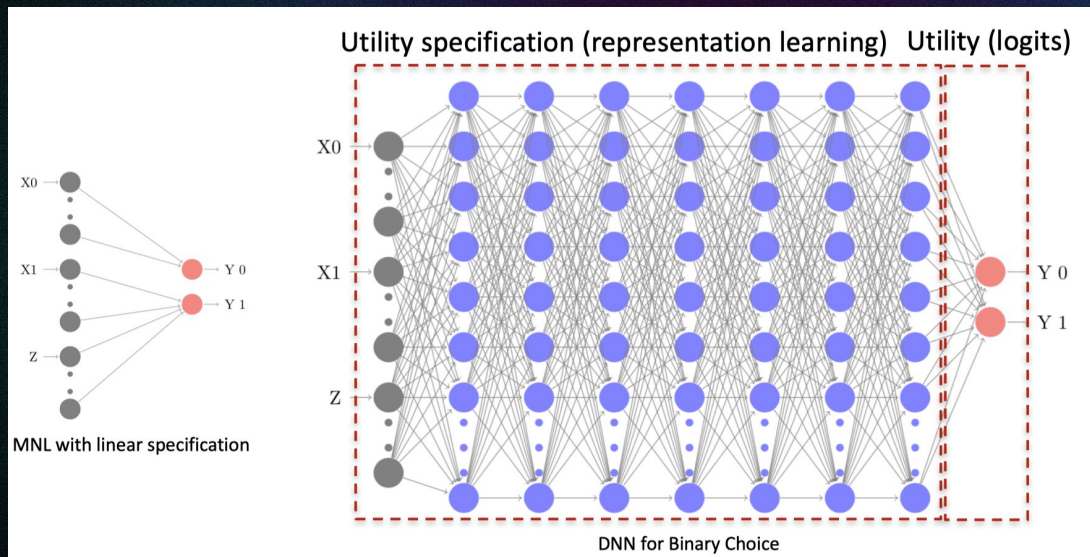
S2: Extracting Economic Information for Interpretation

S3: Architectural Design with Alternative-Specific Utility Function

S4: Theory-Based Residual Neural Network

S5: Deep hybrid model with urban imagery

Extract economic information from DNN in choice analysis



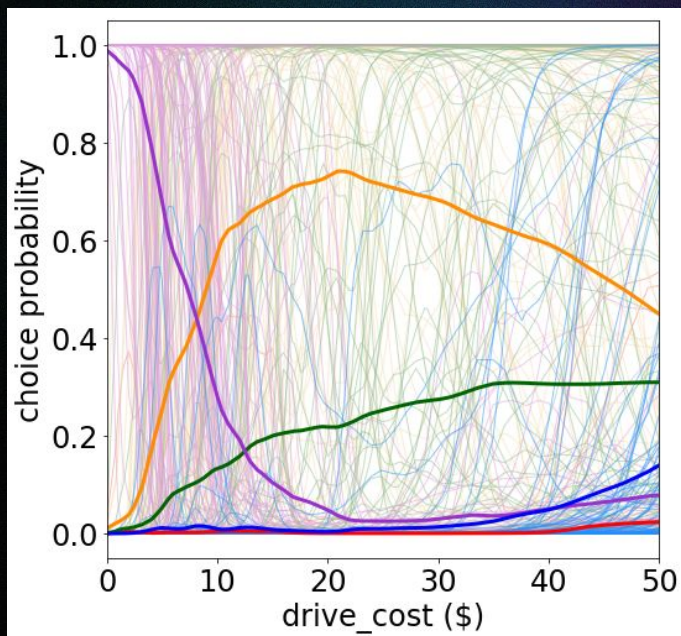
Choice probability function [soft labels]: $\mathbf{s}(\mathbf{x})$
Utility functions [logits]: $\mathbf{V}(\mathbf{x})$

Extract economic information from DNN in choice analysis

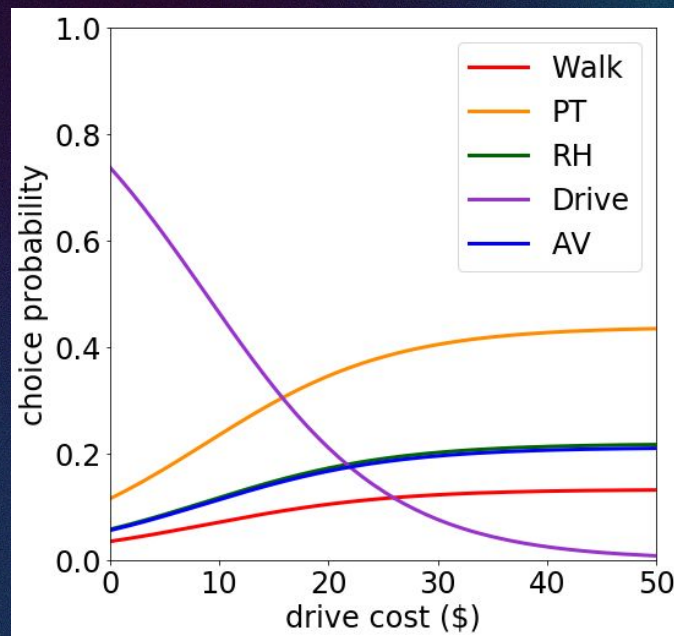
Economic Information	Formula in DNNs
Choice probability	$\hat{s}_k(x_i)$
Choice prediction	$\arg \max_k \hat{s}_k(x_i)$
Market share	$\sum_i \hat{s}_k(x_i)$
Substitution pattern between k_1 and k_2	$\hat{s}_{k_1}(x_i)/\hat{s}_{k_2}(x_i)$
Social welfare	$\sum_i \frac{1}{\alpha_i} \log \left(\sum_{k=1}^K e^{\hat{V}_{ik}} \right) + C$
Change of social welfare	$\sum_i \frac{1}{\alpha_i} \left[\log \left(\sum_{k=1}^K e^{\hat{V}_{ik}^1} \right) - \log \left(\sum_{k=1}^K e^{\hat{V}_{ik}^0} \right) \right]$
Probability derivative of alt. k w.r.t. x_{ij}	$\partial \hat{s}_k(x_i) / \partial x_{ij}$
Elasticity of alt. k w.r.t. x_{ij}	$\frac{\partial \hat{s}_k(x_i)}{\partial x_{ij}} \cdot \frac{x_{ij}}{\hat{s}_k(x_i)}$
MRS between x_{ij_1} and x_{ij_2}	$-\frac{\partial \hat{s}_k(x_i) / \partial x_{ij_1}}{\partial \hat{s}_k(x_i) / \partial x_{ij_2}}$
VOT (x_{ij_1} = time, x_{ij_2} = money)	$-\frac{\partial \hat{s}_k(x_i) / \partial x_{ij_1}}{\partial \hat{s}_k(x_i) / \partial x_{ij_2}}$

Substitution Patterns of Five Alternatives

5L-DNNs



MNL



Architectural Design with Alternative-Specific Utility Function

AI for Transportation

Wang, S., Mo, B., & Zhao, J. (2020). Deep neural networks for choice analysis: Architecture design with alternative-specific utility functions. Transportation Research Part C
<https://doi.org/10.1016/j.trc.2020.01.012>

L1: DNNs and Discrete Choice Models

S1: Background

S2: Extracting Economic Information for Interpretation

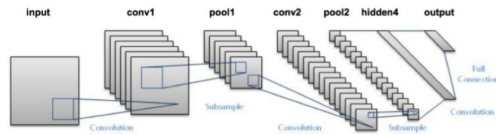
S3: Architectural Design with Alternative-Specific Utility Function

S4: Theory-Based Residual Neural Network

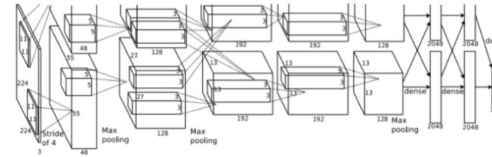
S5: Deep hybrid model with urban imagery

Architectural design of DNNs

LeNet, 1998



AlexNet, 2012



GoogLeNet, 2014

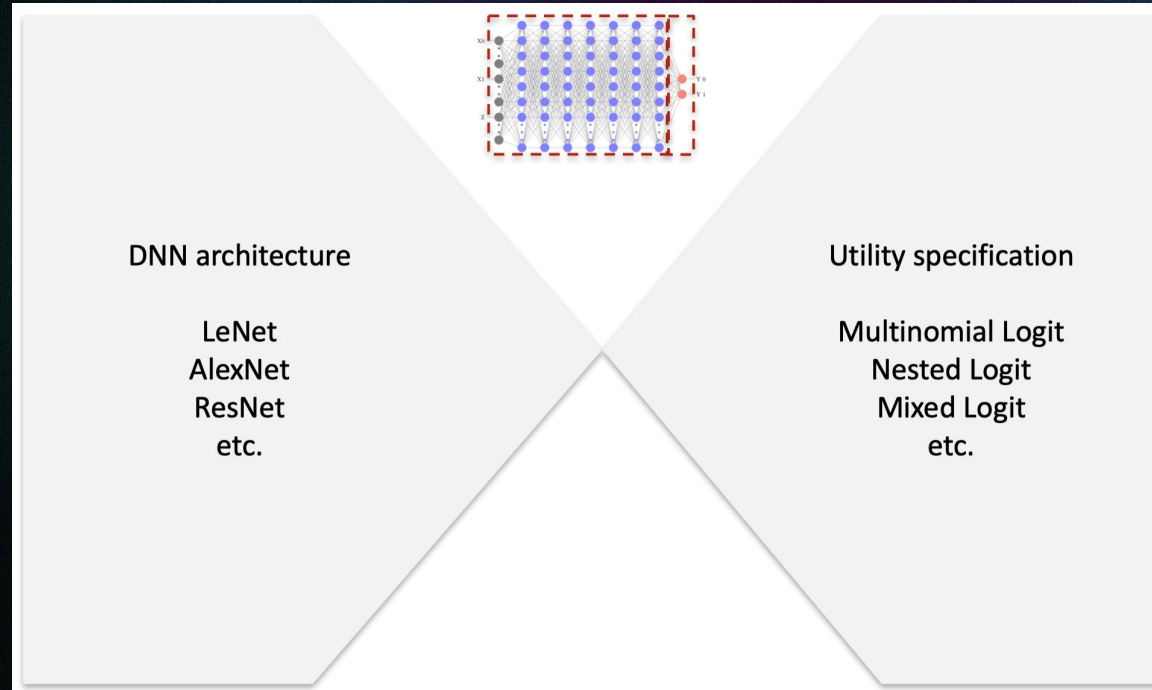


ResNet, 2015

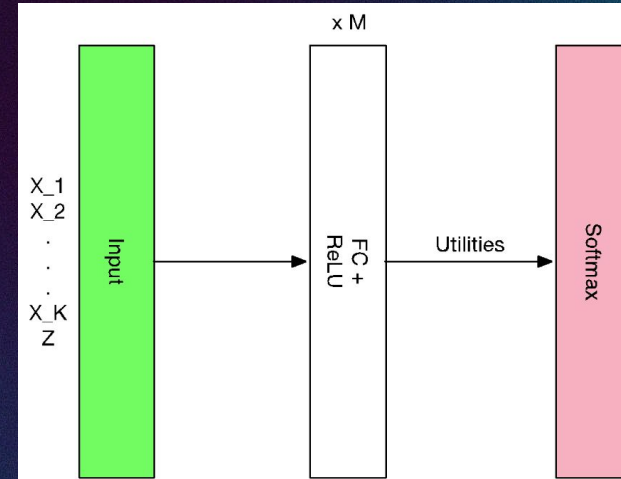
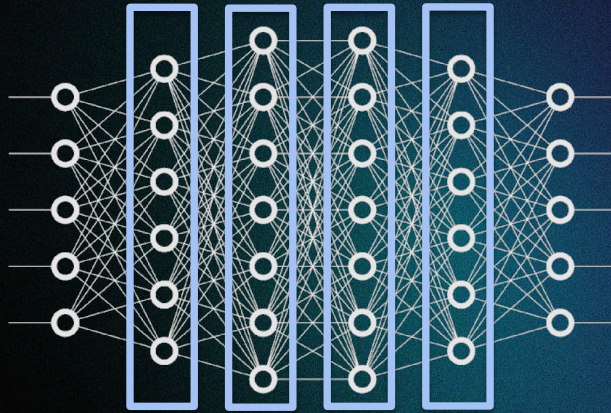


1. Architectural design is critical for DNN performance.
2. However, the DNN is not architecturally interpretable.

Use some utility specification insights to design new DNN architectures for choice analysis



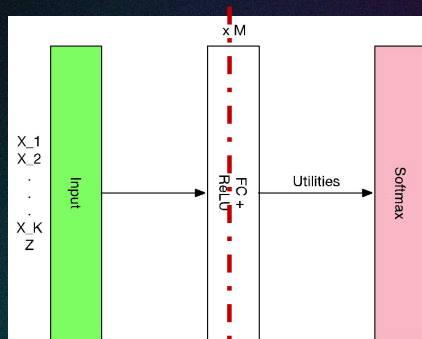
Utility Specification in Fully Connected DNN (F-DNN)



$$V_k = f(x_1, x_2, \dots, x_K, z)$$

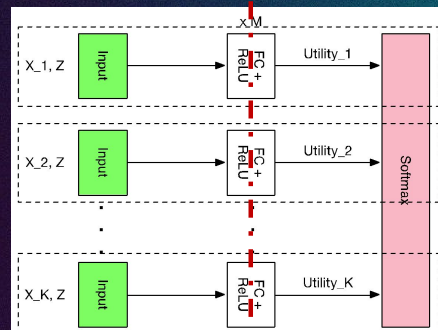
Using Utility Theory to Design DNN

Fully connected DNN (F-DNN)



$$V_k = f(x_1, x_2, \dots, x_K, z)$$

DNN with alternative-specific connection (ASU-DNN)



$$V_k = f(x_k, z)$$

Sparsity as Regularization Introduced by Utility Theory

$$W_{F-DNN} = \begin{bmatrix} \begin{bmatrix} w_{T_1,1} & \dots & w_{T_1,T_1} \\ \vdots & \ddots & \vdots \\ w_{T_1,1} & \dots & w_{T_1,T_1} \end{bmatrix} & \begin{bmatrix} w_{T_1,T_1+1} & \dots & w_{T_1,T_1+T_2} \\ \vdots & \ddots & \vdots \\ w_{T_1,T_1+1} & \dots & w_{T_1,T_1+T_2} \end{bmatrix} & \dots & \begin{bmatrix} w_{T_1,\sum_{i=1}^{K-1} T_i+T_K} & \dots & w_{T_1,\sum_{i=1}^{K-1} T_i+T_K} \\ \vdots & \ddots & \vdots \\ w_{T_1,\sum_{i=1}^{K-1} T_i+T_K} & \dots & w_{T_1,\sum_{i=1}^{K-1} T_i+T_K} \end{bmatrix} \\ \begin{bmatrix} w_{T_1+1,1} & \dots & w_{T_1+1,T_1+T_2} \\ \vdots & \ddots & \vdots \\ w_{T_1+1,1} & \dots & w_{T_1+1,T_1+T_2} \end{bmatrix} & \begin{bmatrix} w_{T_1+1,T_1+1} & \dots & w_{T_1+1,T_1+T_2} \\ \vdots & \ddots & \vdots \\ w_{T_1+1,T_1+1} & \dots & w_{T_1+1,T_1+T_2} \end{bmatrix} & \dots & \begin{bmatrix} w_{T_1+1,\sum_{i=1}^{K-1} T_i+T_K} & \dots & w_{T_1+1,\sum_{i=1}^{K-1} T_i+T_K} \\ \vdots & \ddots & \vdots \\ w_{T_1+1,\sum_{i=1}^{K-1} T_i+T_K} & \dots & w_{T_1+1,\sum_{i=1}^{K-1} T_i+T_K} \end{bmatrix} \\ \vdots & \vdots & \ddots & \vdots \\ \begin{bmatrix} w_{\sum_{i=1}^{K-1} T_i+1,1} & \dots & w_{\sum_{i=1}^{K-1} T_i+1,T_1} \\ \vdots & \ddots & \vdots \\ w_{\sum_{i=1}^{K-1} T_i+1,1} & \dots & w_{\sum_{i=1}^{K-1} T_i+1,T_1} \end{bmatrix} & \begin{bmatrix} w_{\sum_{i=1}^{K-1} T_i+1,T_1+1} & \dots & w_{\sum_{i=1}^{K-1} T_i+1,T_1+T_2} \\ \vdots & \ddots & \vdots \\ w_{\sum_{i=1}^{K-1} T_i+1,T_1+1} & \dots & w_{\sum_{i=1}^{K-1} T_i+1,T_1+T_2} \end{bmatrix} & \dots & \begin{bmatrix} w_{\sum_{i=1}^{K-1} T_i+1,\sum_{i=1}^{K-1} T_i+T_K} & \dots & w_{\sum_{i=1}^{K-1} T_i+1,\sum_{i=1}^{K-1} T_i+T_K} \\ \vdots & \ddots & \vdots \\ w_{\sum_{i=1}^{K-1} T_i+1,\sum_{i=1}^{K-1} T_i+T_K} & \dots & w_{\sum_{i=1}^{K-1} T_i+1,\sum_{i=1}^{K-1} T_i+T_K} \end{bmatrix} \end{bmatrix}$$

Reduce Estimation Error

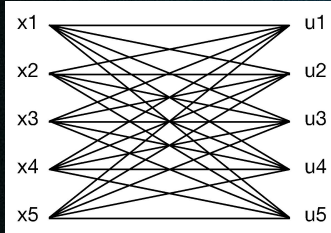
$$\hat{R}_n(F_2|_S) \leq \frac{(\sqrt{2 \log(D)} + 1) \sqrt{\frac{1}{N} \sum_{i=1}^N \|x_i\|^2}}{\sqrt{N}} \times c^D T^D$$

$$W_{ASU-DNN} = \begin{bmatrix} \begin{bmatrix} w_{T_1,1} & \dots & w_{T_1,T_1} \\ \vdots & \ddots & \vdots \\ w_{T_1,1} & \dots & w_{T_1,T_1} \end{bmatrix} & \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} & \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} & \begin{bmatrix} w_{T_1+T_2,T_1+1} & \dots & w_{T_1+T_2,T_1+T_2} \\ \vdots & \ddots & \vdots \\ w_{T_1+T_2,T_1+1} & \dots & w_{T_1+T_2,T_1+T_2} \end{bmatrix} & \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} \\ \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} & \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} & \begin{bmatrix} w_{\sum_{i=1}^{N-1} T_i+1, \sum_{i=1}^{N-1} T_i+1} & \dots & w_{\sum_{i=1}^{N-1} T_i+1, \sum_{i=1}^{N-1} T_i+T_N} \\ \vdots & \ddots & \vdots \\ w_{\sum_{i=1}^{N-1} T_i, \sum_{i=1}^{N-1} T_i+1} & \dots & w_{\sum_{i=1}^{N-1} T_i, \sum_{i=1}^{N-1} T_i+T_N} \end{bmatrix} \end{bmatrix}$$

$$\hat{R}_n(F_1|_S) \leq \frac{(\sqrt{2 \log(D)} + 1) \sqrt{\frac{1}{N} \sum_{i=1}^N \|x_i\|^2}}{\sqrt{N}} \times \frac{c^D T^D}{K^{D/2}}$$

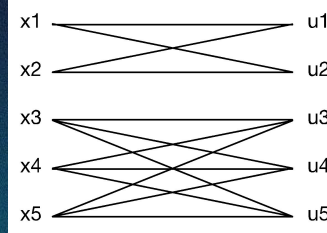
DNN Design with **Utility Connectivity Graph**

F-DNN



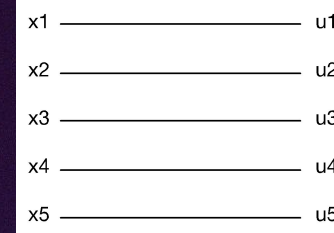
$$A_{F-DNN} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

NSU-DNN



$$A_{NSU-DNN} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

ASU-DNN



$$A_{ASU-DNN} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

1. Implementation in DNN: **Group-LASSO**
2. **Local** vs. **Global** Connectivity Control

Substitution Patterns and Independence of Irrelevant Alternative (IIA) Constraint

ASU-DNN
(IIA)

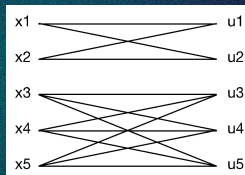


$$P_1/P_3 = f(x_1, x_3; z)$$

$$P_1/P_2 = f(x_1, x_2; z)$$

MNL

NSU-DNN
(IIA in Nests)

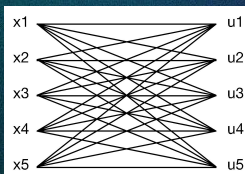


$$P_1/P_2 = f(x_1, x_2; z)$$

$$P_1/P_3 = f(x_1, x_2, \dots, x_5; z)$$

NL

F-DNN



$$P_1/P_3 = f(x_1, x_2, \dots, x_5; z)$$

$$P_1/P_2 = f(x_1, x_2, \dots, x_5; z)$$

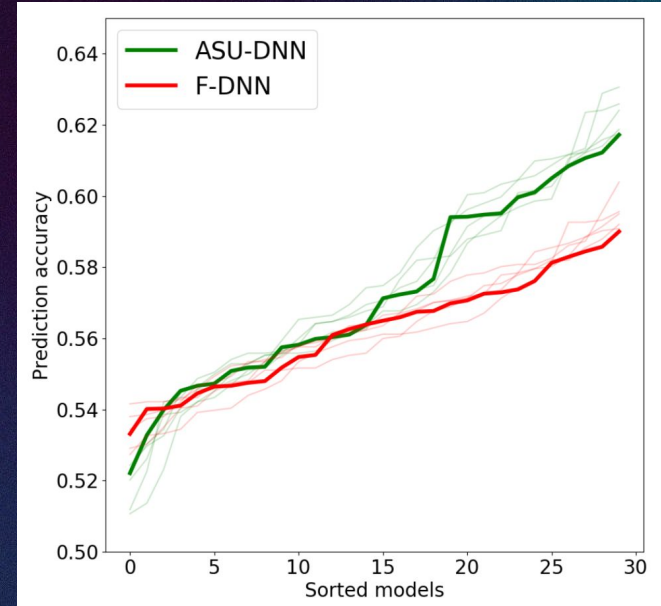
MXL

Experiment Setup

1. Two datasets (SGP & TRAIN; travel mode choice as the target output)
2. Random hyper-parameter searching
3. Sample size. SGP: 8,418; TRAIN: 2,929.
4. Evaluate
 - a. Prediction accuracy
 - b. Effectiveness of regularization
 - c. Interpretation: substitution patterns

Empirical Result 1: **ASU-DNN outperforms F-DNN**

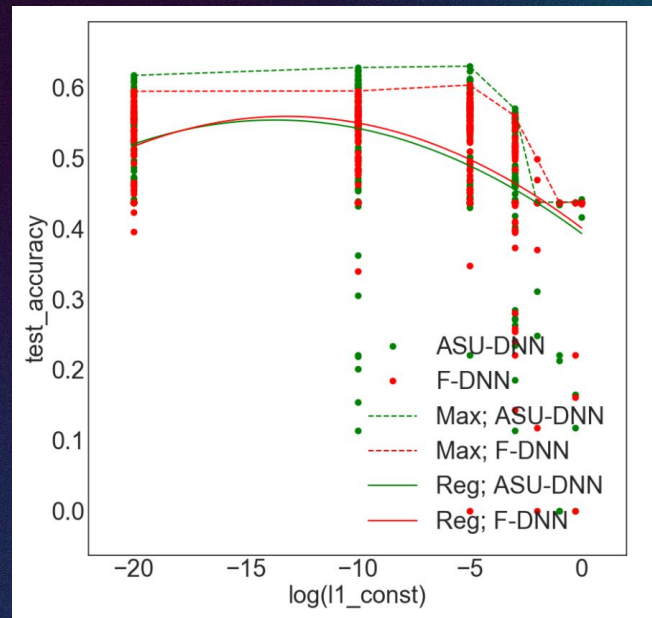
- It holds true for both validation and testing sets
- It holds true for SGP and TRAIN datasets.
- ASU-DNN and F-DNN outperform other nine classifiers



Singapore Testing Set

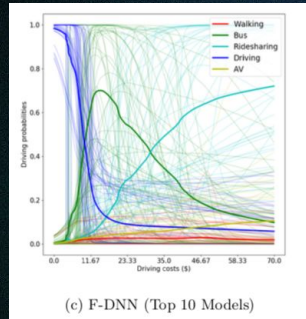
Empirical Result 2: **Alternative-specific connectivity is better than generic-purpose regularizations**

- Explicit regularizations (e.g. L2 regularization)
- Implicit regularizations (e.g. # of iterations)
- Architectural hyper-parameters (depth & width)

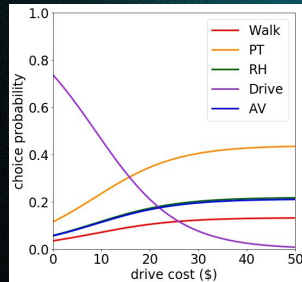


Empirical Result 3: Interpreting the Substitution Pattern

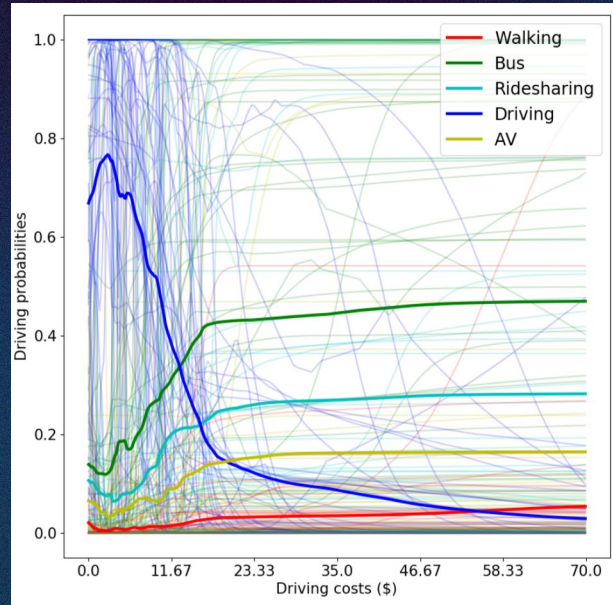
F-DNNs



MNLs



ASU-DNNs



Architectural Design with Alternative-Specific Utility Functions

Feasible & effective to introduce behavioral knowledge into DNN:

- Creating new DNN architecture
- Improving prediction
- Domain-knowledge-based regularization
- Improving interpretation

Theory-Based Residual Neural Network

AI for Transportation

Wang, S., Mo, B., & Zhao, J. (2021). Theory-based residual neural networks: A synergy of discrete choice models and deep neural networks. Transportation research part B <https://doi.org/10.1016/j.trb.2021.03.002>

L1: DNNs and Discrete Choice Models

S1: Background

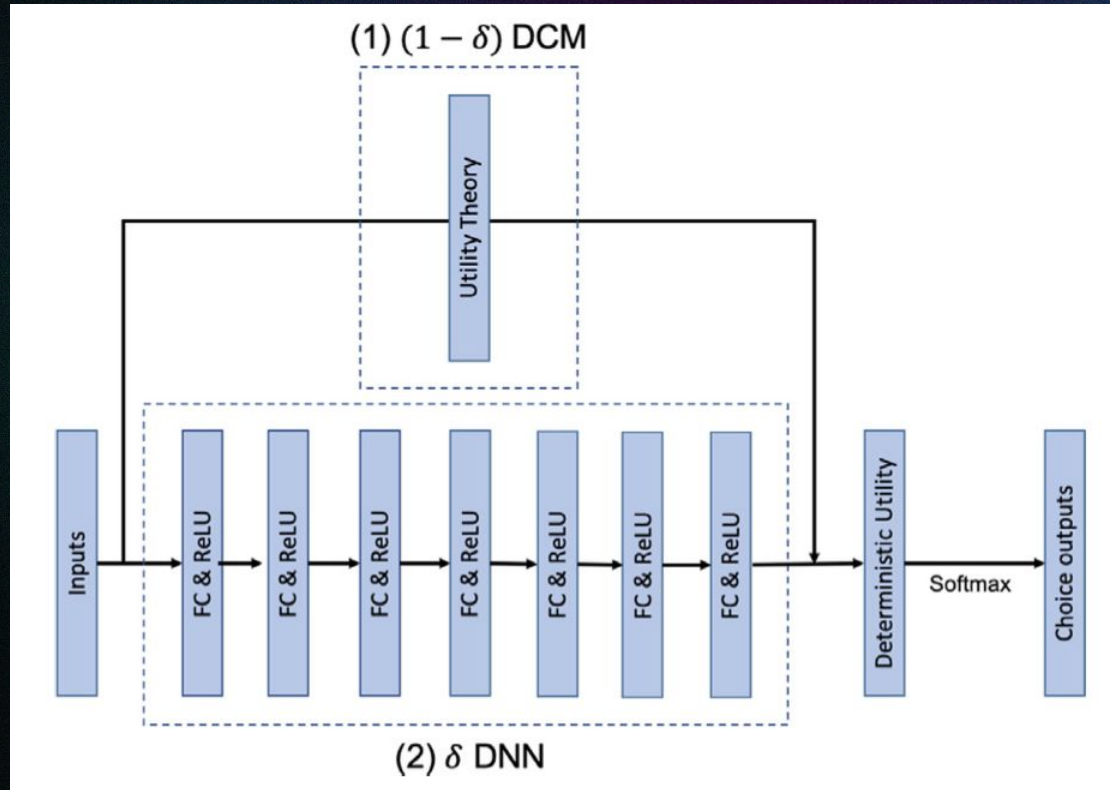
S2: Extracting Economic Information for Interpretation

S3: Architectural Design with Alternative-Specific Utility Function

S4: Theory-Based Residual Neural Network

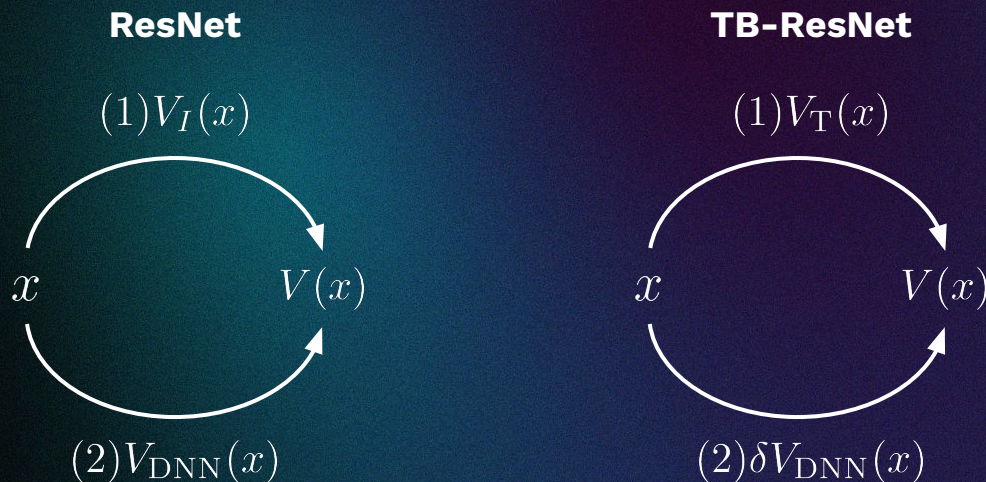
S5: Deep hybrid model with urban imagery

Theory-Based Residual Neural Network (TB-ResNet)



Theory-Based Residual Neural Network (TB-ResNet)

$$\nu_{\text{TB-ResNet},ik} = (1 - \delta) \nu_{T,ik} + \delta \nu_{DNN,ik}$$



Theory-Based Residual Neural Network (TB-ResNet)

$$\nu_{\text{TB-ResNet},ik} = (1 - \delta) \nu_{T,ik} + \delta \nu_{DNN,ik}$$

1. $\delta(\lambda)$
2. Two-stage training: (1) $V_T(x)$ and (2) $V_{DNN}(x)$
3. Generic for any utility maximization framework

Three Instances of Theory-Based Residual Neural Network

Three Instances

$$\nu_{\text{TB-ResNet},ik} = (1 - \delta) \nu_{T,ik} + \delta \nu_{DNN,ik}$$

1. CM-ResNet (choice modeling)
 - o choose between K alternatives
2. PT-ResNet (prospect theory)
 - o choose between two risky payoffs (x, p)
3. HD-ResNet (hyperbolic discounting)
 - o temporal decisions (x, t)

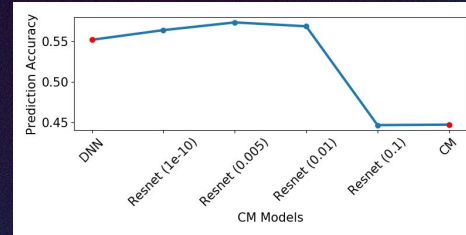
Evaluation Metrics for the Three TB-ResNets

$$\nu_{\text{TB-ResNet},ik} = (1 - \delta) \nu_{T,ik} + \delta \nu_{DNN,ik}$$

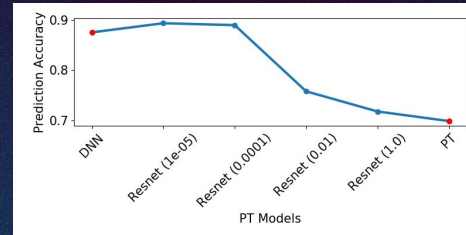
1. Prediction Accuracy
2. Interpretation (local information)
3. Robustness

1. Prediction accuracy

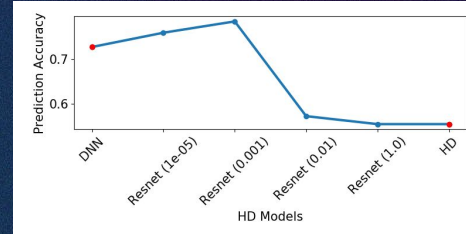
CM Scenario



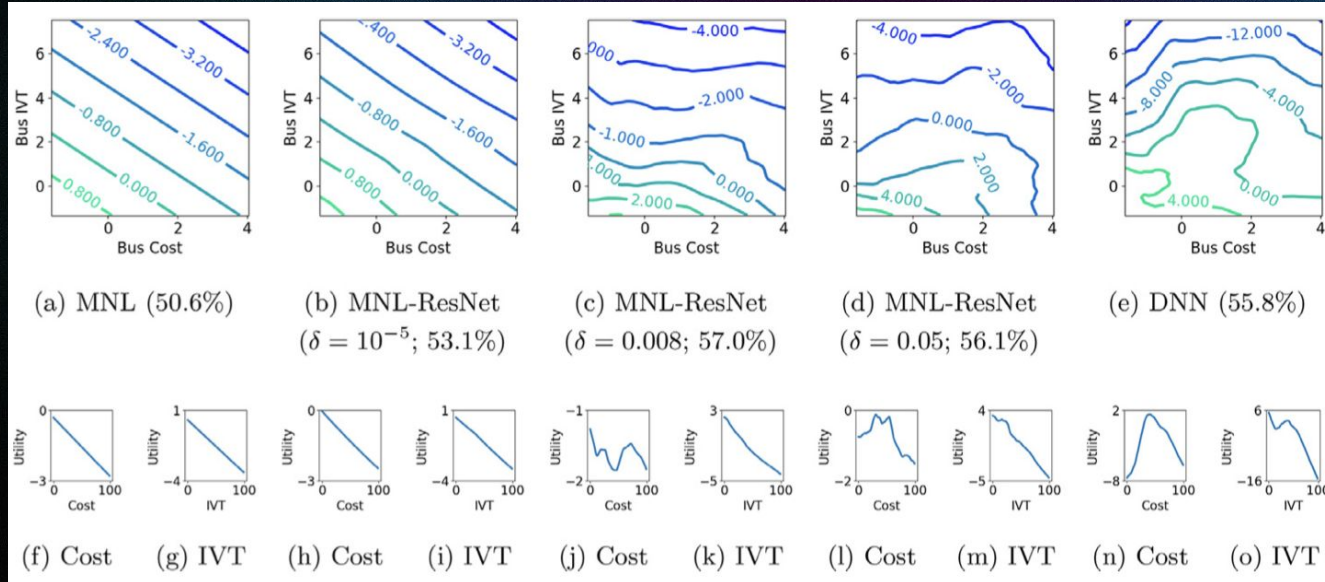
PT Scenario



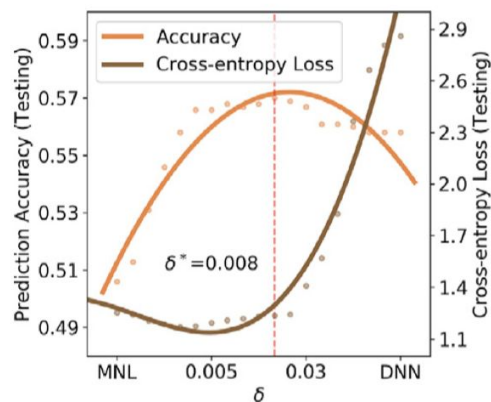
HD Scenario



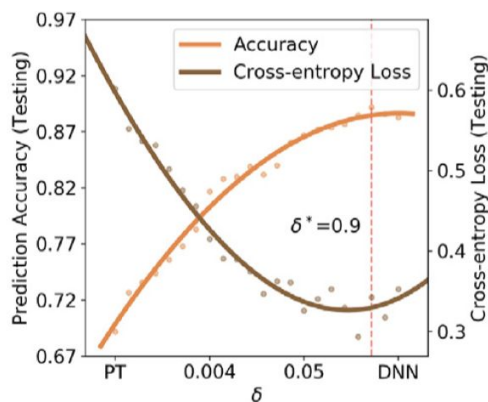
2. Interpretability of utility function in the choice modeling scenario



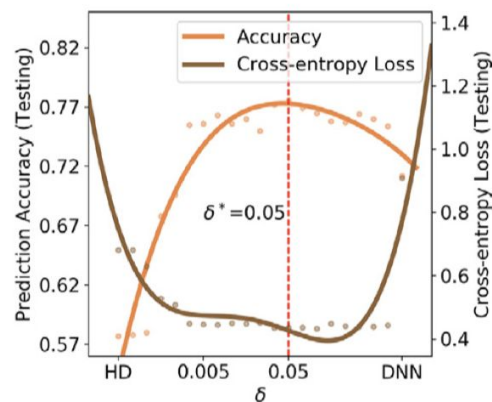
3. Model performance and delta



(a) MNL



(b) PT



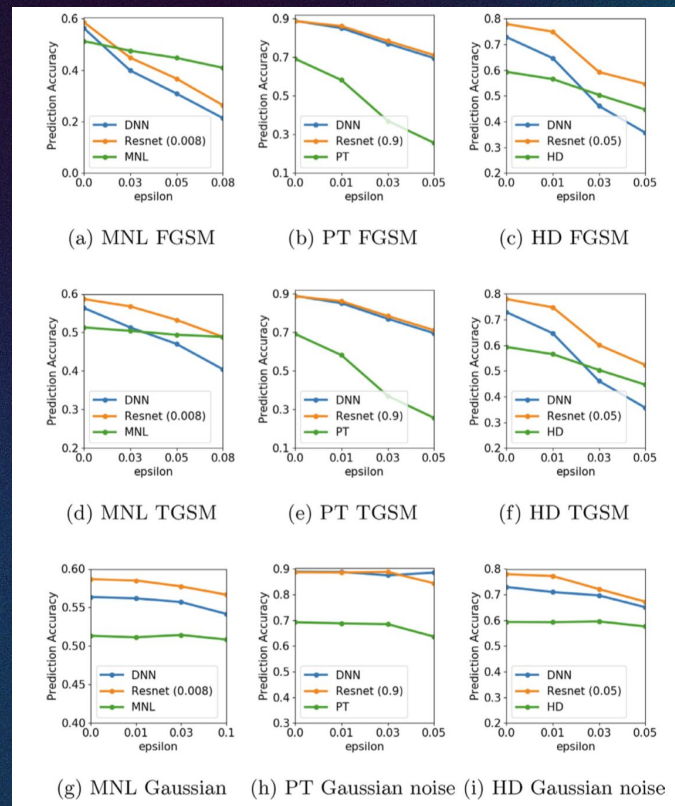
(c) HD

4. Robustness

Fast Gradient Sign Method Attack (FGSM)

Target Gradient Sign Method Attack (TGSM)

Gaussian noise



Comparing TB-ResNet to DCMs and DNNs

Models	Prediction	Interpretability	Robustness
Compared to DNNs	Marginal improvement (by stabilization and regularization)	Significant improvement (by stabilization and regularization)	Significant improvement (by stabilization and regularization)
Compared to DCMs	Significant improvement (by augmenting and enriching utility function)	Significant improvement (by augmenting and enriching utility function)	No improvement

Theory-Based Residual Neural Network

- Analogy to ResNet
- Simple and generic
- Flexible combination of DCMs and DNNs
 - Provide mutual benefits
- Advantages:
 - Higher prediction accuracy
 - Better interpretability
 - Robust to adversarial attacks

Deep Hybrid Model With Urban Imagery

AI for Transportation

L1: DNNs and Discrete Choice Models

S1: Background

S2: Extracting Economic Information for Interpretation

S3: Architectural Design with Alternative-Specific Utility Function

S4: Theory-Based Residual Neural Network

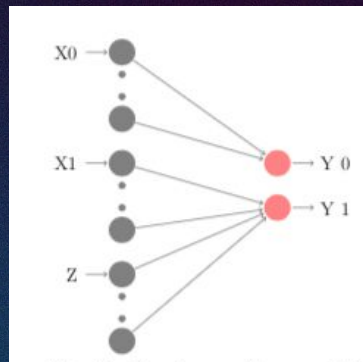
S5: Deep hybrid model with urban imagery

Demand modeling vs. deep learning

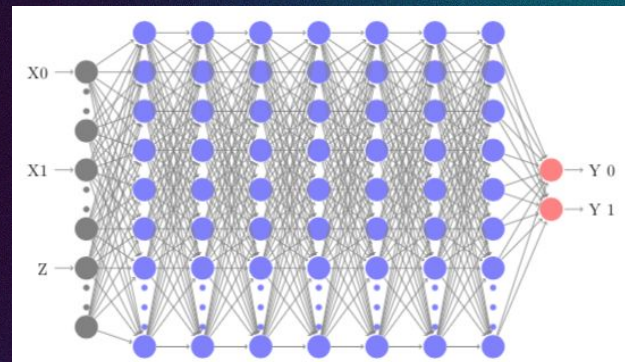
How to learn from both sides?

- Deep choice models

Demand modeling



Deep learning



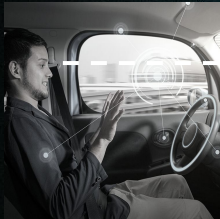
Tension

- **Data:** structured vs. unstructured data
- **Model:** shallow vs. deep model
- **Representation:** handcrafted vs. automatic learning
- **Common sense:** interpretation vs. prediction

1 - Imagery is ubiquitous in human decisions

Quote: "Not everything that can be counted counts.
Not everything that counts can be counted."

Driver decisions



(\$50K/year, male, 25 years old, etc.)



Street-view imagery

Consumer decisions



(\$65K/year, female, 33 years old, etc.)

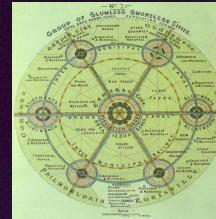


Product imagery

2 - Imagery has economic values

Quote: "A picture is worth a thousand words"

Social benefits of urban planning

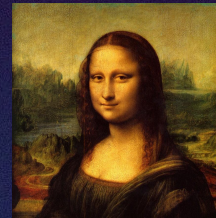


Garden city – health



Radiant city – efficiency

\$ value of painting



Mona lisa
>\$900M



The Starry Night
>\$100M

Allegory of the Cities

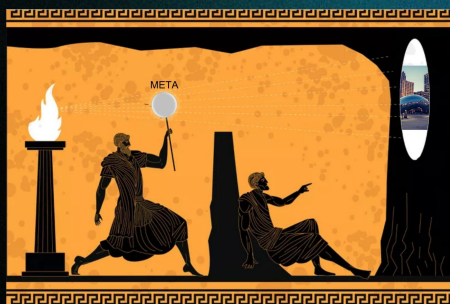
Cities projected to **numbers**



Cities projected to **graphs**



Cities projected to **images**



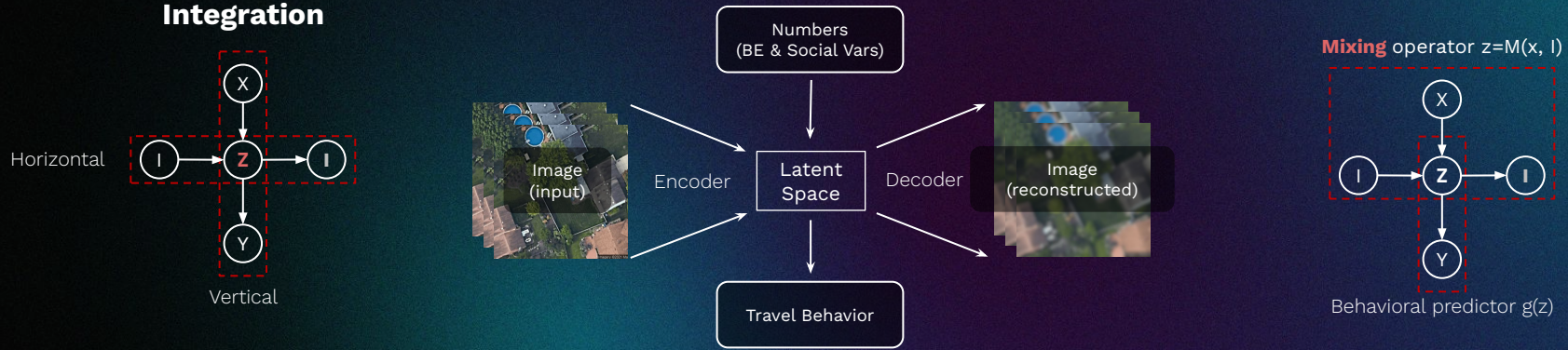
Cities projected to **natural language**



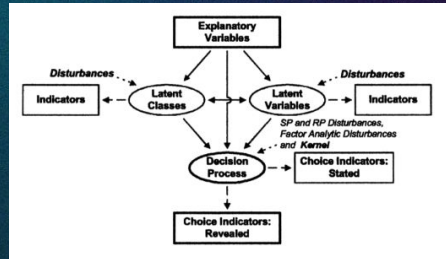
How to integrate numeric data and urban imagery

1. improve travel behavioral analysis
2. create economic interpretation for urban imagery

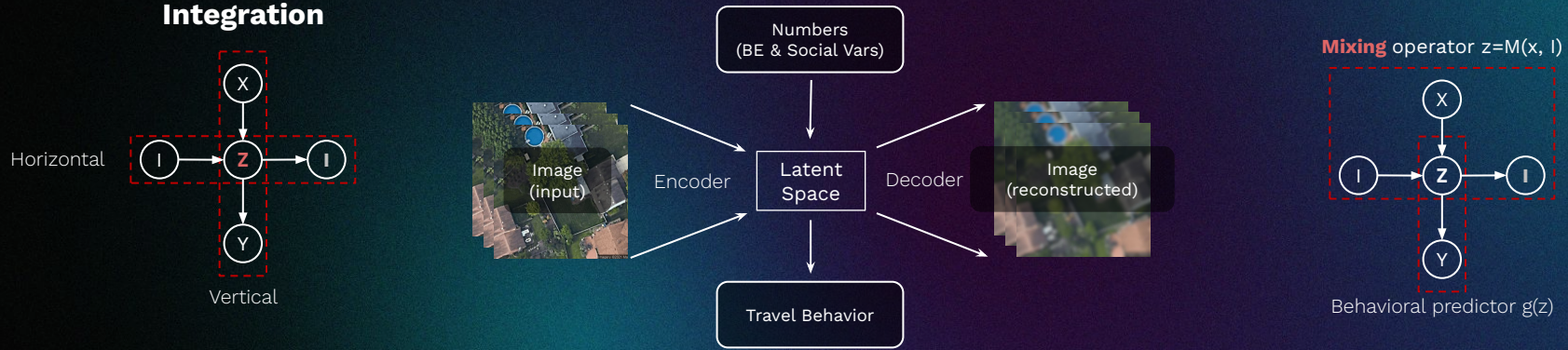
Deep Hybrid Model



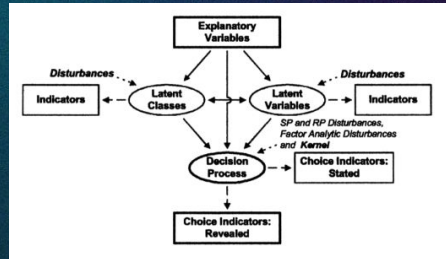
Similarity to hybrid demand model



Deep Hybrid Model



Similarity to hybrid demand model



Contextualizing the research

Variables

- Socio-demographics (x); urban imagery (I); travel behavioral output (y)
- $\text{Dim}(x) = 10$; $\text{Dim}(I) = 2048$; $\text{Dim}(y) = 1$ or 3

Context - Chicago

- Travel survey + satellite imagery

Tasks

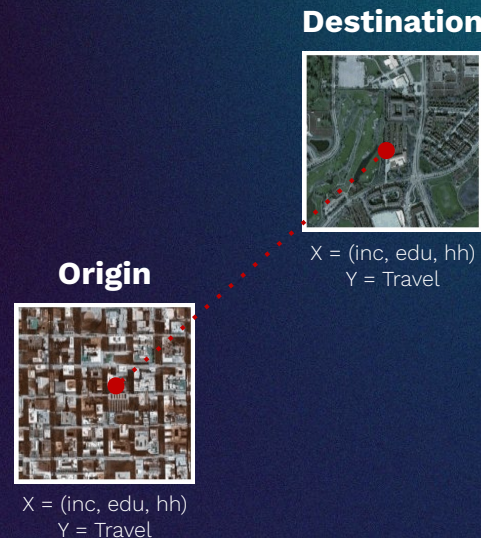
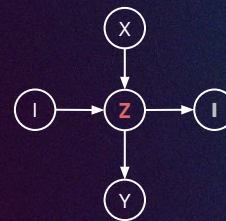
- Aggregate travel mode choices: node-feature prediction
- Disaggregate travel mode choices: edge-feature prediction

Model - DHM

- $y = g(\mathbf{M}(x, I))$
- Mixing operator $\mathbf{M}$: $\lambda \in [0, 1]$ as a hyperparameter
- Behavioral predictor g : $\theta \in [0, 1]$ as a hyperparameter; LASSO and RIDGE

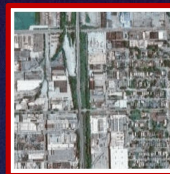
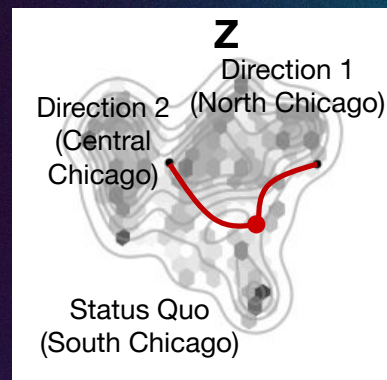
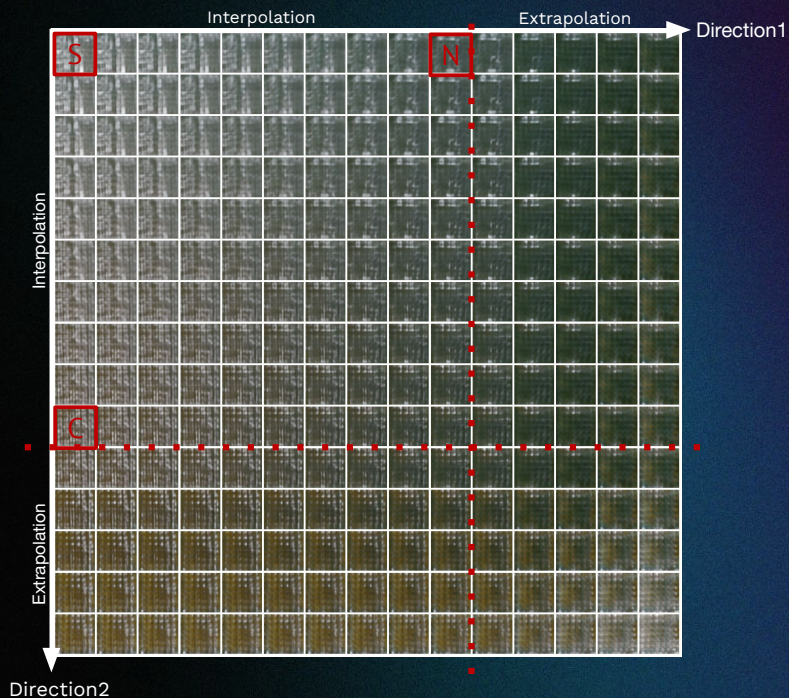
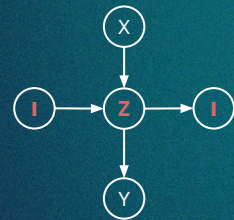
Metrics

- R2, KL, and LL

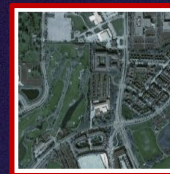


Economic interpretation of urban imagery: navigating the imagery space

Interpolation and extrapolation
generate NEW urban design (225 | 100 | 3)



Status Quo
(South Chicago)



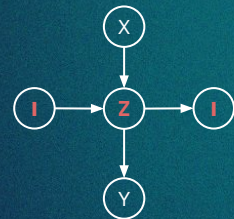
Target 1
(North Chicago)



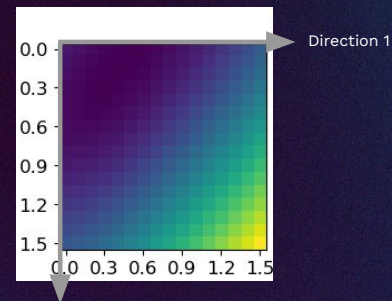
Target 2
(Central Chicago)

Social welfare and behavioral changes associated with urban imagery

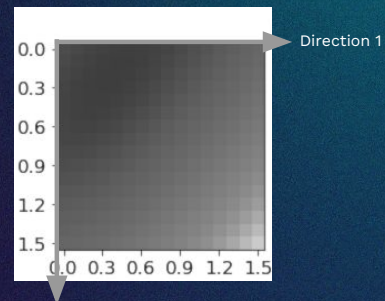
Interpolation and extrapolation
generate NEW urban design (225 | 100 | 3)



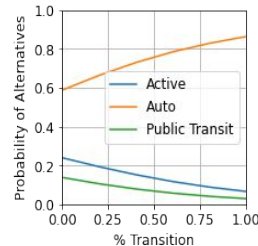
Social Welfare Map (X)



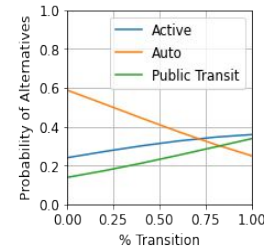
\$ Value Map (X)



Y change (Direction 1)



Y change (Direction 2)



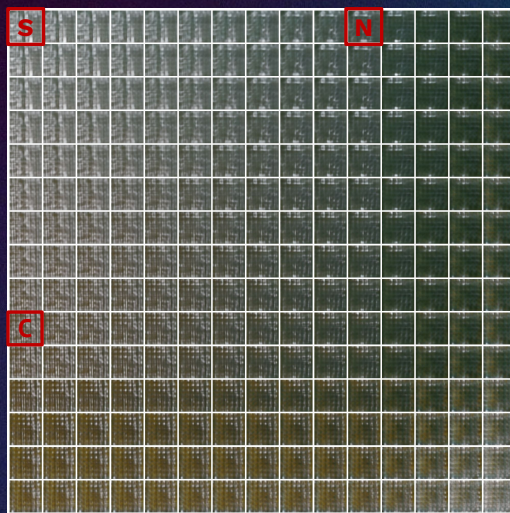
Implication 1: Demand Modeling

Deep hybrid model

- Generalized hybrid demand model
- Incorporating Imagery
- Behavioral realism
- Economic interpretation of imagery

Implication 2: Urban Planning

Automatic and socially-aware urban planning



AI for Transportation

L1: DNNs and Discrete Choice Models

S1: Background

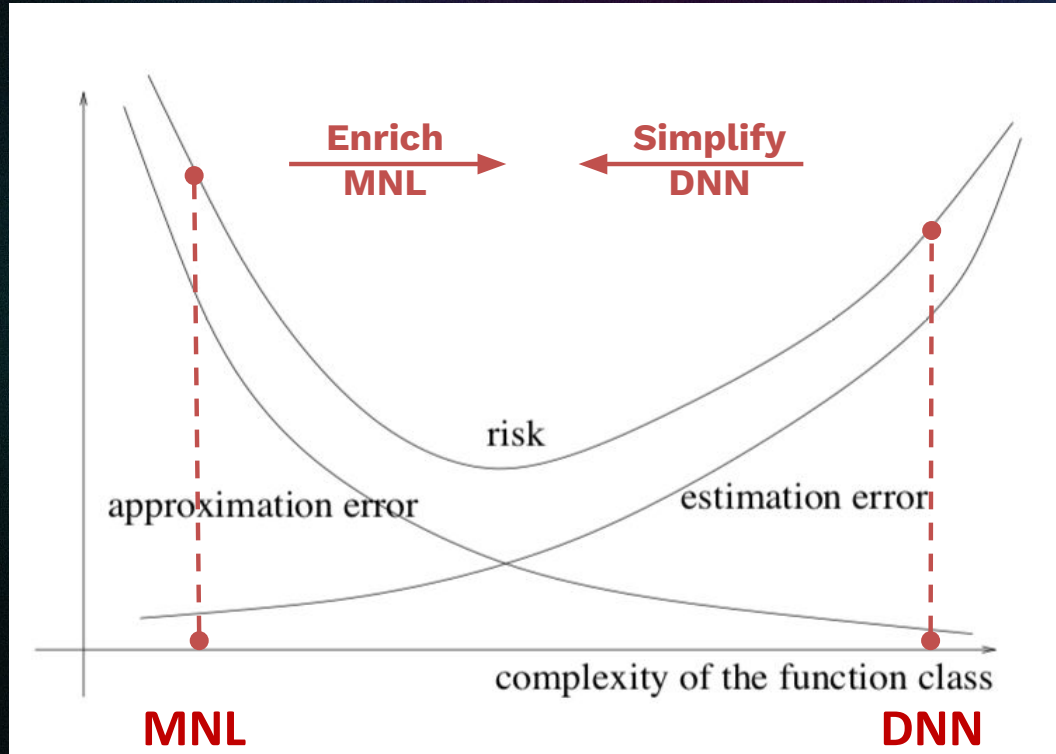
S2: Extracting Economic Information for Interpretation

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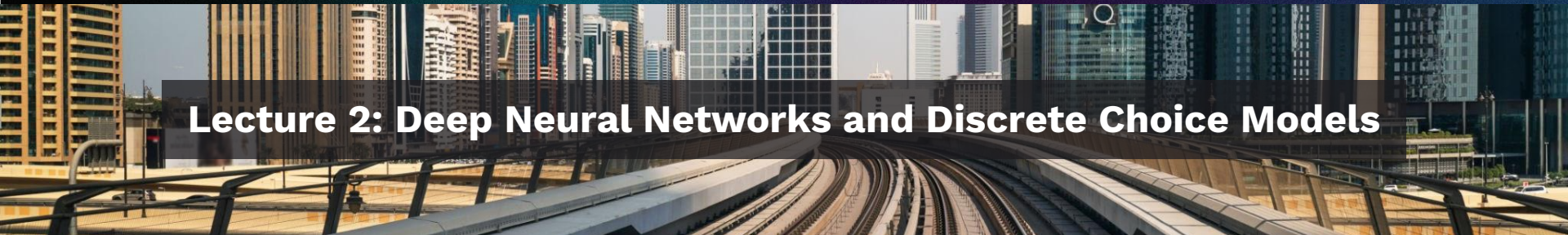
Prediction Errors = Approximation + Estimation Errors



How to address the problems by using **utility theory**?

Challenges	Theoretical Foundations	Solutions
High sensitivity to hyper-parameters	Statistical challenge: balance approximation and estimation errors	1. Regularization methods: model ensemble (Krizhevsky et al. 2012), data augmentation, dropouts (Hinton et al. 2012), early stopping; 2. Architectural design: AlexNet (Krizhevsky, et al. 2012), GoogleNet (Szegedy et al. 2015), and ResNet (He et al. 2016) 3. Automatic hyperparameter tuning: Bayesian neural networks (Snoek 2012, Snoek 2015), or reinforcement learning (Zoph 2016, Zoph 2017) 4. Statistical learning theory (Vapnik 2013, Bartlett 2002, Bartlett 2017, Neyshabur 2015, Golowich 2017)
Model non-identification	Optimization challenge: find global optimum	New perspective: it does not matter...
Local irregularities	Robustness challenge: stabilize local information	1. Robust training methods: adversarial training (Kurakin et al. 2016), defensive knowledge distillation (Papernot et al. 2016), mini-max robust training (Madry et al. 2017) 2. Monotonicity constraints: (Gupta et al. 2016)

AI for Transportation: From concepts to implementation



Lecture 2: Deep Neural Networks and Discrete Choice Models